ADC-aware PTQ and Post-ADC Low-Rank Correction for Llama

Analog in-memory compute · Post-training quantization

Alina Burykina

Constructor University — April 2026

Optical GPU for LLM Inference

Photonic tensor core for Transformer linear layers

LLM inference is mostly dense linear algebra

QKV projections · FFN · Output projection

$\sim 10^3\times$
energy-efficiency gain

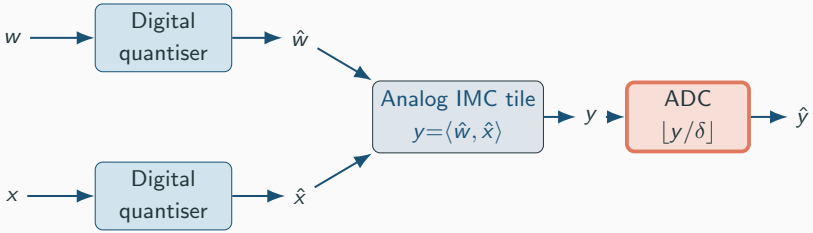
$\leq 10^{17}$ **ops/s**
projected inference throughput

≤ 100 **GHz**
optical input switching

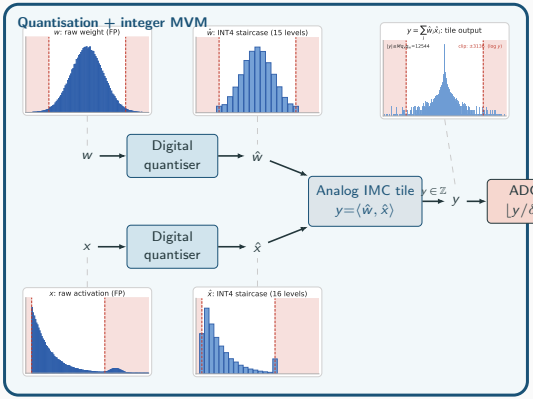
Not a GPU replacement — a specialized photonic accelerator for Transformer linear layers.[1]

[1] Bandyopadhyay et al., *Single chip photonic deep neural network with accelerated training*, arXiv:2308.05226

Analog MVM Signal Path

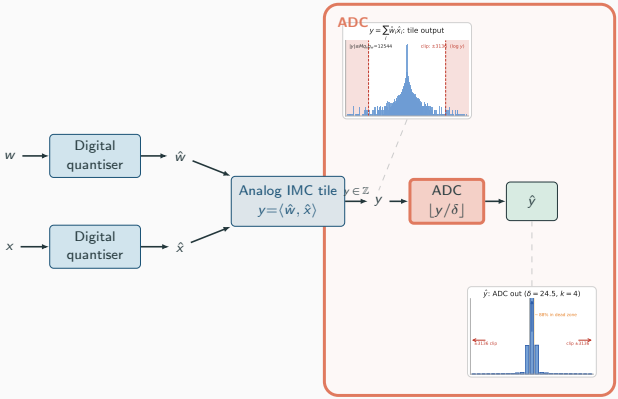


Digital Quantisation



Quantise: $\hat{v} = \mathrm{clip}(\lfloor v/s \rceil,\ -q,\ q),\ s = \max|v|/q,\ q = 2^{b-1}-1$ **Dequantise:** $\tilde{v} = s \cdot \hat{v}$

ADC: Reading the Analog Accumulation



ADC read: $\hat{y} = \mathrm{clip}\big(\lfloor y/\delta \rfloor,\ -2^{b_a-1},\ 2^{b_a-1}-1\big)$ **Step:** $\delta = \dfrac{2M\, q_x\, q_w}{2^{b_a}\, k}$ **Dead zone:** $|y| < \delta \ \Rightarrow\ \hat{y} = 0$

Research Questions

Q1 Can we **reshape** x, w so that ADC-quantised forward matches FP behaviour *without* retraining the model?
→ *Method I: FlatQuant + trainable diagonals*

Q2 How low can **pure PTQ** drive $\mathrm{PPL}_{\mathrm{ADC}}$ on INT8?
→ *block-wise propagated calibration*

Q3 Does the same recipe work on **INT4 ADC**, where δ is 330× smaller?
→ *staged training, α-mix loss*

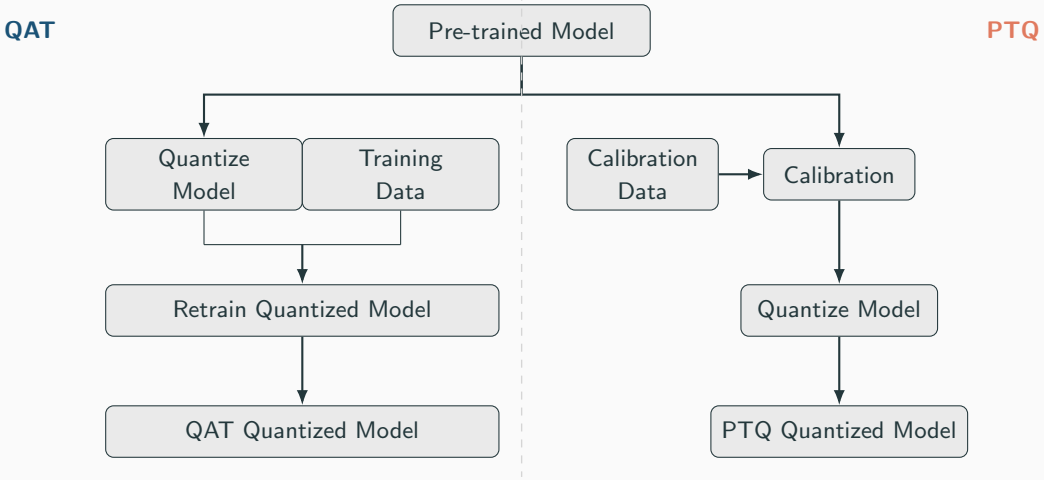
Q4 When PTQ **plateaus**, what is the minimum intervention beyond PTQ?
→ *Method II: Post-ADC residual LoRA*

Thesis in one line

The right place to correct ADC error is **after** *the ADC, not before it.*

Setup: Llama-3.2-1B, WikiText-2 for calibration, WikiText-2 + C4 for evaluation, ADC spec b_a=8, k=16, M=256.

QAT vs PTQ


QAT
Pre-trained Model
PTQ
Quantize Model
Training Data
Calibration Data
Calibration
Retrain Quantized Model
Quantize Model
QAT Quantized Model
PTQ Quantized Model

FlatQuant: Rotate, Flatten, Calibrate per Layer

1. Linear invariance.

For any invertible T:

$$y = xW^\top = \underbrace{(xT^{-1})}_{\tilde{x}}\underbrace{(TW^\top)}_{\tilde{W}^\top}.$$

Choose T so $\tilde{x}$ is *flat* $\Rightarrow$ integer quantisation & ADC coverage improve.

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2. Kronecker factorisation.

Parameterise $T \in \mathbb{R}^{d\times d}$ as

$$T = T_1 \otimes T_2, \quad T_1 \in \mathbb{R}^{\sqrt{d}\times\sqrt{d}},\ T_2 \in \mathbb{R}^{\sqrt{d}\times\sqrt{d}}$$

params $\mathcal{O}(d)$ instead of $\mathcal{O}(d^2)$.

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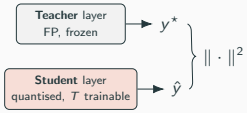
3. Teacher-student per-layer calibration.

For each linear layer L:

- **Teacher** (FP): $y^\star = L_{\mathrm{FP}}(x^\star)$
- **Student** (quantised + T): $\hat{y} = \hat{L}_T(\hat{x})$
- Minimise $\|\hat{y} - y^\star\|^2$ over T's only.



Our Extension I: Block-wise Calibration & Propagation

From per-projection to per-block.

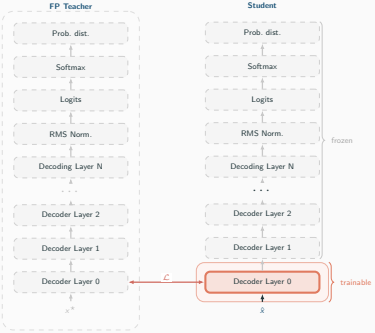
We reconstruct the *whole transformer block*:

$$\mathcal{L} \;=\; \left\| \hat{B}(\hat{x}) - B_{\mathrm{FP}}(x^{\star}) \right\|_2^2$$

Each block is trained *one at a time* — here, Decoder Layer 0.

Propagation.

Input to block ℓ is the ADC-quantised output of block $\ell-1$.



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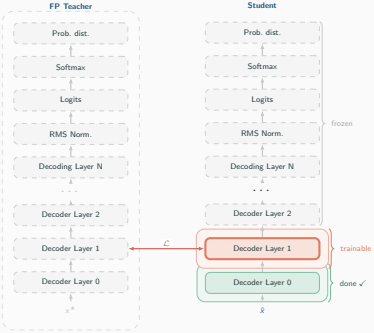
Now **Decoder Layer 1**. Its input is the ADC-quantised output of Layer 0.

Propagation.

Error accumulates *the same way at train time and at inference*.



Our Extension I: α-Mixed Objective

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FlatQuant calibrates each linear layer separately (MSE on its output). We reconstruct the *whole transformer block*:

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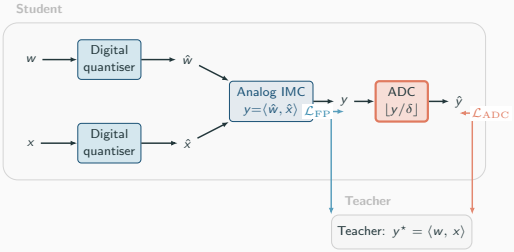
Gradients flow through q/k/v → attn → o → gate/up → down.

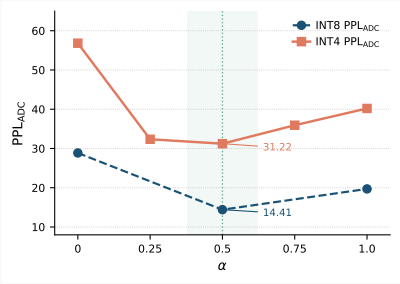
α-mixed objective.

Feed both FP and ADC inputs through the student block in one step:

$$\mathcal{L}(\alpha) \;=\; (1-\alpha)\,\mathcal{L}_{\mathrm{FP}} \;+\; \alpha\,\mathcal{L}_{\mathrm{ADC}}$$

α=0: clean — easy to fit, ignores ADC noise.
α=1: realistic — matches inference but hard to optimise (noisy grads).
α=0.5: *both* signals at once.





Our Extension I: Block-wise Calibration & α-Mix

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Gradients flow through q/k/v → attn → o → gate/up → down, so ADC error from earlier projections is seen by later ones during optimisation.

Propagation across blocks.

Input to block ℓ is the ADC-quantised output of block $\ell-1$, *not* its FP teacher activation. The error accumulates *the same way at train time and at inference*.

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α	INT8 $\mathrm{PPL}_{\mathrm{ADC}}$	INT4 $\mathrm{PPL}_{\mathrm{ADC}}$
0.0	28.86	38.5
0.5	**14.41**	**27.56**
1.0	19.70	31.8

What this buys us (INT8)

$\mathrm{PPL}_{\mathrm{ADC}}$ 28.86 → **14.41** just from switching the loss from per-projection MSE to block α-mix. No extra parameters.

Our Extension II: Trainable Diagonals (`add_diag`)

Motivation. A rotation T flattens the *shape* of the activation cloud, but not its *per-channel scale*. On INT4 ADC this leaves residual outliers that shove a few channels into the clipping region.

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Fix. Absorb a trainable diagonal $D = \mathrm{diag}(d_1, \ldots, d_c)$ into the rotation:

$$\tilde{T} \;=\; T \cdot D, \qquad d_i \in [10^{-4}, 10]$$

Invariance preserved — equal and opposite scaling on the next layer:

$$y \;=\; (xD^{-1}T^{-1})\,(TDW^{\top}).$$

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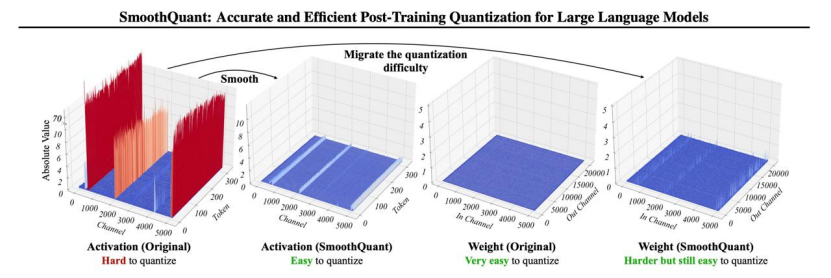
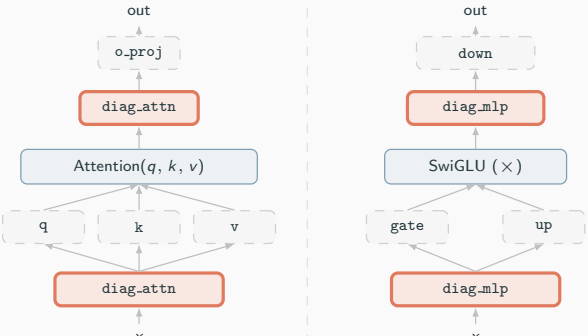


Our Extension II: Trainable Diagonals (add_diag)

Where we place them.

- diag_attn: before q/k/v/o projections
  (absorbs attention-head variance)
- diag_mlp: before gate/up/down projections
  (absorbs FFN outlier channels — SwiGLU)



Ablation (Llama-3.2-1B, w4a4, ADC PPL↓):

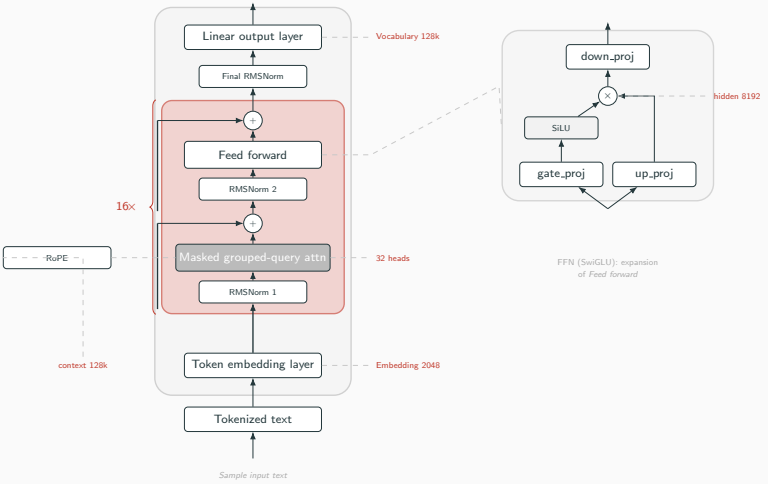
FlatQuant +	$\mathrm{PPL}_{\mathrm{bypass}}$	$\mathrm{PPL}_{\mathrm{ADC}}$	dead-rate
— (base)	13.17	2355	80.9%
diag_mlp only	12.58	1090	45.2%
diag_attn only	12.89	830	32.1%
both diagonals	**12.04**	**420**	**10.4%**

(numbers to be re-filled from v3 logs)

Why it matters for ADC

Diagonals absorb the per-channel dynamic range — so a single δ can serve every channel. This is the only PTQ knob we have that directly targets the *fixed-step* nature of ADC.

Staged Training: What Is Learnt, When



Per-block params

7 Kron. rotations $T_{q,k,v,o,g,u,d}$; 3 MLP diagonals $D^{\mathrm{m}}_{g,u}$; 3 attn diagonals $D^{\mathrm{a}}_{q,k,v}$.

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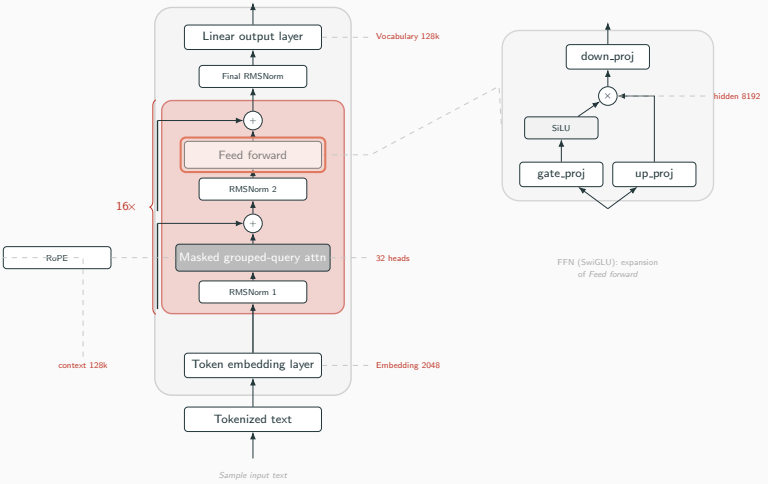

Stage A | MLP-first

Freeze attention. Train $T_{g,u,d}$ and $D^{m}_{g,u}$ with α=0 — clean forward warmup.

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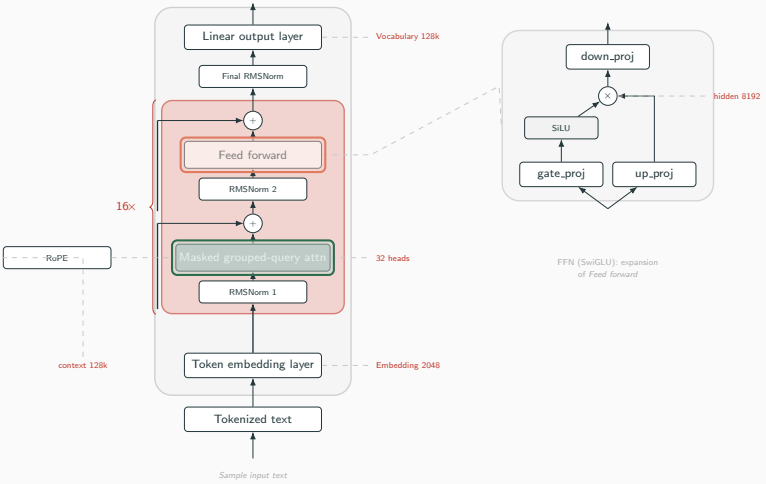
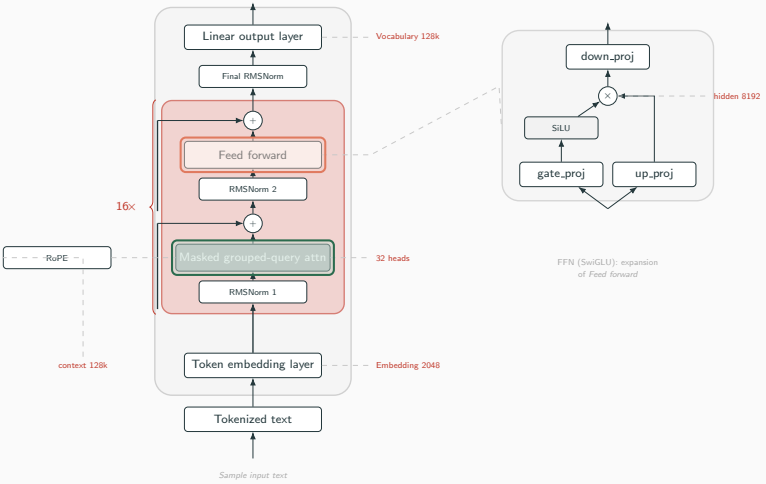


Stage B | full block

Unfreeze attention. Add $T_{q,k,v,o}$ and $D^{a}_{q,k,v}$. Turn on α=0.5.

Staged Training: What Is Learnt, When



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Why staged?

Joint α=0.5 from scratch diverges — attention gradients dominate before rotations align. MLP warmup $\Rightarrow$ $\sim$2.3$\times$ better INT4 ADC PPL.

TODO: $\times$2.3 diverges — README staged vs joint $\times$1.03 ADC PPL, $\times$2.3

PTQ Results: INT8 and INT4 ADC, WikiText-2

Method	bits (w / a)	$\mathrm{PPL}_{\mathrm{bypass}}$	$\mathrm{PPL}_{\mathrm{ADC}}$	dead-rate	notes
FP baseline	16 / 16	10.53	10.53	—	reference
FlatQuant per-proj MSE	8 / 8	**10.01**	28.86	10.3%	base implementation
+ block calib., α=0.5	8 / 8	11.00	**14.41**	10.3%	INT8 PTQ best
FlatQuant per-proj MSE	4 / 4	13.17	2355	80.9%	dead zone explodes
+ diag_mlp	4 / 4	12.58	1090	45.2%	MLP outliers tamed
+ diag_attn	4 / 4	12.04	420	12.1%	attn outliers tamed
+ block calib., α=0.5	4 / 4	20.23	38.5	10.4%	single-stage
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Take-aways on PTQ alone

- INT8: block+α-mix is the single biggest lever — no extra params. $2\times$ **better**.
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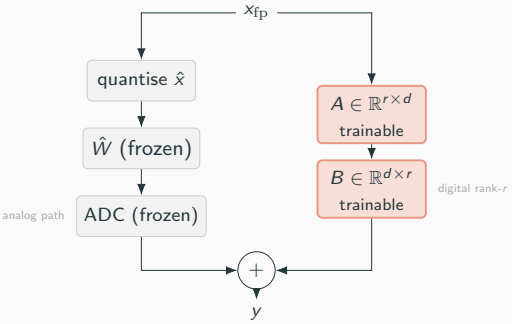
The plateau

Every PTQ lever combined still leaves a 17 *PPL gap* on INT4 (27.6 vs 10.5 FP). PTQ by itself has plateaued.
→ *Method II: Post-ADC LoRA.*

Method II: Post-ADC Residual LoRA

Key idea. Freeze everything PTQ produced. Add a tiny low-rank branch that sits in parallel with the ADC-quantised projection, reads the FP input, and adds to the ADC output:

$$y = \underbrace{\mathrm{ADC}_{\mathrm{frozen}}(\hat{x})}_{\text{quantised path}} + \underbrace{\frac{\alpha_{\mathrm{LoRA}}}{r} B\big(A(x_{\mathrm{fp}})\big)}_{\text{residual correction, rank } r}$$



Loss: distill from FP teacher.

$$\mathcal{L}_{\mathrm{LoRA}} = \mathrm{CE}(\hat{p}, y) + \lambda\, \mathrm{KL}(\hat{p} \,\|\, p_{\mathrm{FP}})$$

λ=1, α_{LoRA}/r scaling as in the LoRA paper.

Analogous to the original LoRA figure (Hu et al. 2021), but added **after** *the ADC, not to pre-trained weights.*

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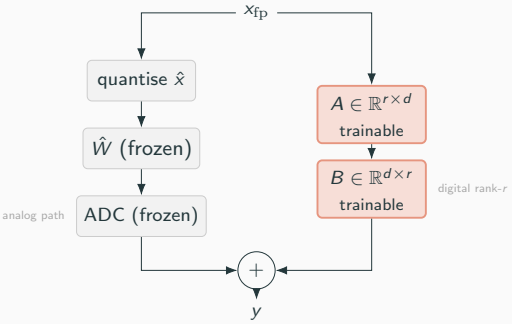

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What is trained:

- Only A, B for each of 7 projections (q, k, v, o, gate, up, down).
- Everything else (weights, rotations, diagonals, ADC params) is frozen.
- FP teacher runs once to produce p_{FP}; no gradient through it.

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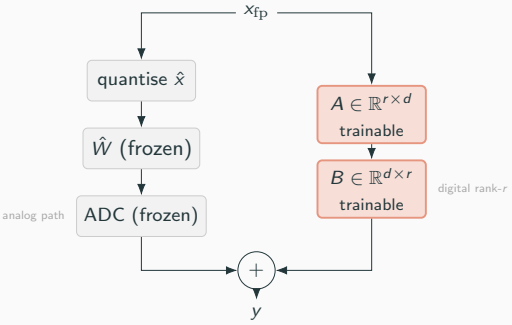

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Why post-ADC works

Pre-ADC LoRA *fails* — its correction gets quantised by the fixed δ. Post-ADC, the correction is added in the digital domain with FP precision.

LoRA Ablations and Overhead

Ablations (Llama-3.2-1B, INT4, $\mathrm{PPL}_{\mathrm{ADC}}\downarrow$)

Axis	Setting	$\mathrm{PPL}_{\mathrm{ADC}}$
Rank r	r=1	15.8
	r=2	14.5
	r=4	**13.96**
	r=8	14.02
Loss	CE only	14.6
	CE + KL	**13.96**
Layers	last 8 only	16.8
	all 16	**13.96**
Placement	Pre-ADC	28.4 (fails)
	Post-ADC	**13.96**

Overhead on Llama-3.2-1B (d=2048, 16 blocks)

total LoRA params (r=4)	$\sim$1.8 M
as fraction of FP weights	**0.15%**
FP-teacher forward	1$\times$ cached
student forward + back	$\sim$5$\times$ FP MAC
training time (1 GPU, 5 ep.)	$\sim$45 min
inference MAC overhead	$<$ 2%

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Loss	CE only	14.6
	CE + KL	**13.96**
Layers	last 8 only	16.8
	all 16	**13.96**
Placement	Pre-ADC	28.4 (fails)
	Post-ADC	**13.96**

Overhead on Llama-3.2-1B (d=2048, 16 blocks)

total LoRA params (r=4)	$\sim$1.8 M
as fraction of FP weights	**0.15%**
FP-teacher forward	1$\times$ cached
student forward + back	$\sim$5$\times$ FP MAC
training time (1 GPU, 5 ep.)	$\sim$45 min
inference MAC overhead	$<$ 2%

Observations

- r=4 saturates — adding capacity doesn't help.
- KL adds 0.6 PPL improvement for free.
- Early layers matter most (biggest ADC error reservoir).
- Pre-ADC placement fails because the correction gets quantised.

Final Results: INT4+LoRA Beats INT8 PTQ

Configuration	bits	WikiText-2 $\mathrm{PPL}_{\mathrm{ADC}}$	C4 $\mathrm{PPL}_{\mathrm{ADC}}$	extra params
FP baseline	16/16	10.53	12.10	—
INT8 — FlatQuant per-proj	8/8	28.86	33.41	0
INT8 — block+α-mix **(our PTQ)**	8/8	14.41	16.52	0
INT4 — FlatQuant per-proj	4/4	2355	3180	0
INT4 — staged+α-mix+diag **(our PTQ)**	4/4	27.60	31.18	0
INT4 + Post-ADC LoRA **(ours, final)**	4/4	**13.96**	**15.80**	0.15%

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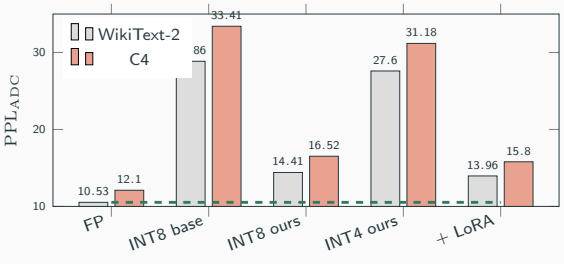
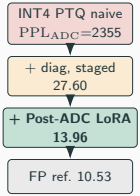


Headlines

- **INT4 + LoRA (13.96) beats best INT8 PTQ (14.41)** using half the bits.
- Gap to FP shrinks from 17.1 PPL (INT4 PTQ) to 3.4 PPL.
- Generalises: C4 pattern mirrors WikiText-2 despite calibration only on WikiText-2.
- Only 0.15% extra parameters; <2% inference overhead.

Takeaways

1. **ADC $\neq$ conventional quantisation.** Fixed δ, fixed clip, no per-layer knobs. Algorithmic fixes must act on x, w statistics.
2. **LLM outliers make a single k infeasible.** Diagonals + rotations solve this at the projection level.
3. **Block-wise + α-mix is the biggest PTQ lever.** Same knob halves INT8 ADC PPL and enables INT4 recovery.
4. **Staged training.** MLP-first warm-up makes INT4 PTQ trainable at all.
5. **Post-ADC LoRA breaks the PTQ plateau.** 0.15% extra params $\rightarrow$ INT4 matches INT8.



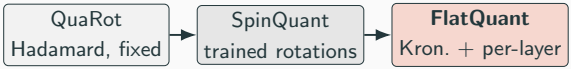
Punchline

The right place to correct ADC error is **after** *the ADC, not before it.*

Backup: Rotation Line of Work

The rotation line of work.

Insert an orthogonal T so that $WT^{-1} \cdot Tx = Wx$ and Tx is easier to quantise:



Why FlatQuant for our work:

- *trainable* — we can add objectives (ADC-aware);
- *Kronecker* ($T = T_1 \otimes T_2$) — memory & compute $\mathcal{O}(\sqrt{d})$;
- per-projection rotations fold naturally into block-wise calibration.

Why ADC is Hard for LLMs: the Outlier Dilemma

LLM activations are heavy-tailed. A tiny fraction of channels carry magnitudes 10–100× larger than the rest *(Dettmers 2022; Xiao 2023)*.

TODO figure

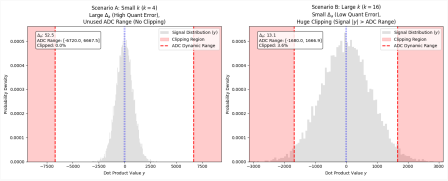
`figs/outlier_hist.pdf`

Activation histogram of a Llama proj.
log-y scale, one heavy tail at $\sim 50\sigma$

Llama-3.2-1B, `q_proj` input, calib set

Why we can't just pick a good k:

Large k (fine δ)	**Small k (coarse δ)**
outliers saturate the ADC range $\Rightarrow$ heavy *clipping*	outliers stay in range
model loses critical channels	dynamic range wasted on outliers
good ADC SNR on bulk	bulk collapses into *dead zone*
bad downstream accuracy	bad ADC SNR everywhere



Takeaway

Hardware-fixed δ + heavy-tailed activations $\Rightarrow$ **no single k works**. The algorithm must reshape x and w *before* they hit the array so that y lands in the ADC's usable range.

Backup: Complete INT8 Results (v2)

Setup: Llama-3.2-1B, w8a8, 128 calibration samples. Testing two ideas on top of baseline FlatQuant: (1) *bin-centre loss* — pushes y_{int} toward ADC bin centres during PTQ; (2) *propagated calibration* — each block sees ADC-quantised inputs from the previous block, matching inference conditions. Key finding: propagation gives a 2× ADC PPL drop; bin-centre loss alone gives nothing.

Experiment	Description	$\text{PPL}_{\text{bypass}}$	PPL_{ADC}	dead_rate
baseline	FlatQuant w8a8	10.01	28.86	10.3%
center mid	bin-centre $\lambda = 0.1$	10.01	28.86	10.3%
prop	propagated calibration	11.01	**14.55**	10.3%
prop+center	propagated + bin-centre	11.00	**14.41**	10.3%

Backup: Complete INT4 v2 Alpha-Sweep

Setup: Llama-3.2-1B, w4a4, 1024 samples. Sweeping α in the mixed loss $\mathcal{L}(\alpha) = (1-\alpha)\mathcal{L}_{\text{FP}} + \alpha\mathcal{L}_{\text{ADC}}$. α=0 = clean FP inputs only; α=1 = ADC-noisy inputs only. Gap = ADC PPL / bypass PPL. Last two rows add diagonals and staged training on top of α=0.5.

Experiment	α	$\text{PPL}_{\text{bypass}}$	PPL_{ADC}	gap
baseline_1024	0.0	19.80	56.83	×2.87
prop_full_1024	1.0	30.32	40.20	×1.32
propalpha_075_1024	0.75	27.46	35.92	×1.31
propalpha_05_1024	0.5	24.24	**31.22**	**×1.29**
propalpha_025_1024	0.25	21.65	32.35	×1.49
add_diag + α=0.5	0.5	20.23	**27.56**	×1.36
staged_mlp_attn	0.5	**18.50**	**27.60**	×1.49

Backup: Complete v7 LoRA Ablation

Setup: Post-ADC LoRA correction on frozen PTQ model (Llama-3.2-1B, w4a4, staged PTQ base: bypass=17.89, ADC=27.55). LoRA adapters trained with CE loss on wikitext2, optionally + KL divergence term. Ablating: target projections (`down+o` vs all 7), rank (1–8), loss (CE vs CE+KL), layer selection (first/last 8), and pre- vs post-ADC placement (pre-ADC fails catastrophically).

Experiment	Targets	Loss	Rank	$\mathrm{PPL}_{\mathrm{bypass}}$	$\mathrm{PPL}_{\mathrm{ADC}}$
base (no LoRA)	—	—	—	17.89	27.55
r1_down_o	down+o	CE	1	—	15.90
r2_down_o	down+o	CE	2	—	15.60
r4_down_o	down+o	CE	4	18.68	15.57
r8_down_o	down+o	CE	8	—	15.24
r4_down_o_ce_kl	down+o	CE+KL	4	19.54	**14.33**
r4_down_o_first8	down+o	CE	4	18.42	16.64
r4_down_o_last8	down+o	CE	4	18.62	20.08
pre_adc_r4_down_o	down+o	CE	4	123.34	**105.63**
r4_all_ce	all 7	CE	4	17.62	16.68
r4_all_ce_kl	all 7	CE+KL	4	19.13	**14.03**

Backup: v8 Generalisation to C4

Setup: Same PTQ base as v7, LoRA trained on wikitext2 only, evaluated on both wikitext2 and C4. Checks whether the best v7 config (`r4_all_ce_kl`) generalises out-of-domain. Without LoRA, the base model has a large C4 ADC gap (46.40 vs bypass 29.41); LoRA closes it to 23.30 on C4 — confirming the correction is not overfitting to the training distribution.

Experiment	Targets	Wiki bypass	Wiki ADC	C4 bypass	C4 ADC
base (no LoRA)	—	18.49	27.26	29.41	46.40
rank8_down_o (CE)	down+o	18.51	15.47	30.48	29.39
r4_down_o_ce_kl	down+o	19.05	**14.27**	29.37	**23.88**
r4_all_ce_kl	all 7	18.70	**13.96**	29.10	**23.30**

Backup: v9 Data Efficiency (lora_nsamples sweep)

Setup: Best config (`r4_all_ce_kl`, `mvm_limit=256`), PTQ fixed at 1024 samples. Varying how many calibration sequences LoRA sees during fine-tuning (64 → 1024). Shows LoRA improves even with 64 samples, but ADC PPL keeps improving up to 1024 — more data helps consistently.

Experiment	nsamples	Wiki $\mathrm{PPL}_{\mathrm{bypass}}$	Wiki $\mathrm{PPL}_{\mathrm{ADC}}$	C4 $\mathrm{PPL}_{\mathrm{bypass}}$	C4 $\mathrm{PPL}_{\mathrm{ADC}}$
r4_all_ce_kl_n64	64	18.42	21.07	29.50	35.62
r4_all_ce_kl_n128	128	18.21	18.09	28.97	30.19
r4_all_ce_kl_n256	256	18.66	15.88	28.84	26.13
r4_all_ce_kl_n512	512	17.99	**14.82**	28.20	**24.75**
r4_all_ce_kl_n1024	1024	18.92	**14.06**	28.79	**23.56**

Config: `r4_all_ce_kl`, `mvm_limit=256`. PTQ fixed at 1024 samples.

Backup: v9 KL Weight × Temperature Sweep

Setup: Base `r4_all_ce_kl` (1024 samples). KL term $\lambda_{\mathrm{KL}} \cdot D_{\mathrm{KL}}(\hat{p}_{\mathrm{ADC}} \,\|\, \hat{p}_{\mathrm{FP}})$ with softmax temperature T. Higher T softens logits before KL, making the teacher signal smoother. λ_{KL}=0.5, T=1 gives the best C4 ADC PPL (23.10); very high λ (2.0/T=4) destabilises training.

Experiment	λ_{KL}	T	Wiki $\mathrm{PPL}_{\mathrm{bypass}}$	Wiki $\mathrm{PPL}_{\mathrm{ADC}}$	C4 $\mathrm{PPL}_{\mathrm{bypass}}$	C4 $\mathrm{PPL}_{\mathrm{ADC}}$
r4_kl025_t1	0.25	1.0	18.77	**13.84**	29.16	**23.23**
r4_kl025_t2	0.25	2.0	19.15	14.07	29.03	23.33
r4_kl05_t1	0.50	1.0	18.92	**13.79**	29.69	**23.10**
r4_kl05_t2	0.50	2.0	18.43	14.05	28.54	23.37
r4_kl10_t2	1.00	2.0	18.90	13.94	29.47	23.17
r4_kl20_t4	2.00	4.0		*failed*		

Base: `r4_all_ce_kl` (1024 samples). Best: $\lambda = 0.5$, $T = 1$ — C4 ADC 23.10.

Backup: Stochastic Propagation (v5)

Setup: Instead of a fixed α, sample α_t each batch — either Bernoulli($\{0, 1\}$) or Beta$(2, 2)$ (concentrated around 0.5). Bernoulli with staged training gives the best bypass PPL ever seen (16.94) but a bad ADC gap, because FP-only batches teach nothing about ADC noise. Beta$(2, 2) \approx$ deterministic α=0.5 in expectation and achieves the best ADC PPL (27.82).

Experiment	Mode	$\mathrm{PPL}_{\mathrm{bypass}}$	$\mathrm{PPL}_{\mathrm{ADC}}$	gap
stoch_bern (flat)	Bernoulli	19.25	35.38	$\times$1.84
stoch_beta2 (flat)	Beta(2,2)	22.90	31.40	$\times$1.37
staged + Bernoulli	Bernoulli	**16.94**	32.44	$\times$1.92
staged + Beta(2,2)	Beta(2,2)	17.16	**27.82**	$\times$1.62
deterministic α=0.5	—	24.24	31.22	$\times$1.29

Bernoulli gives best bypass but worst ADC gap (hard 0/1 switching provides no ADC signal on FP batches). Beta(2,2) $\approx$ deterministic α=0.5 in expectation.

Backup: Hardware Config & Delta Formula

Parameter	Value
b_w (weight bits)	4 / 8
b_x (act. bits)	4 / 8
b_a (ADC bits)	8
`mvm_limit`	256 / 1024
k (ADC parallelism)	16
δ (mvm=256, INT8)	≈ 2016
δ (mvm=256, INT4)	≈ 6.12
δ (mvm=1024, INT8)	≈ 8064
FlatQuant epochs	30
FlatQuant LR	0.005
Cal. samples (INT4)	1024
Cal. samples (INT8)	128

Delta formula ($q = 2^{b-1}-1$):

$$\delta = \frac{2 \cdot \texttt{tile\_in} \cdot q_x \cdot q_w}{2^{b_a} \cdot k}$$

- INT8 (q=127, mvm=256): $\frac{2 \cdot 256 \cdot 127^2}{256 \cdot 16} \approx 2016$
- INT4 (q=7, mvm=256): $\frac{2 \cdot 256 \cdot 7^2}{256 \cdot 16} \approx 6.12$
- INT8, mvm=1024: $\frac{2 \cdot 1024 \cdot 127^2}{256 \cdot 16} \approx 8064$

INT8: useful ADC bins in $\pm\sigma_z \approx 15$–30 (empirical) — explains why INT8 PTQ alone plateaus.

This Work in Context

Method	Key idea	Gap / limit
FlatQuant	Learned Kronecker-decomposed orthogonal transforms per layer	FP loss only; no ADC signal
OmniQuant / LET	Learnable per-channel equivalent scaling (diagonal)	No ADC-aware objective
QEP / prop. calib.	Feed quantised activations into the next layer's calibration	INT8 only; no mixed-loss
PACT / QDrop	Stochastic / clamped quant-aware training	Full re-training required
This work	ADC-aware PTQ (α-mixed, staged diag) + post-ADC LoRA	—

Key novelty

No prior method directly targets the **post-ADC residual** — our LoRA correction is the first to do so.

Post-ADC LoRA: Parameter and Compute Overhead

Trainable parameter count (r=4, all 7 projections):

Projection	A	B
q_proj	4×2048	2048×4
k_proj	4×2048	256×4
v_proj	4×2048	256×4
o_proj	4×2048	2048×4
gate_proj	4×2048	8192×4
up_proj	4×2048	8192×4
down_proj	4×8192	2048×4

Inference overhead (per forward pass):

- Extra FLOPs per projection: $d_{\text{in}}\cdot r + r\cdot d_{\text{out}} \ll d_{\text{in}}\cdot d_{\text{out}}$
- Example: `q_proj`: 8192+8192 vs. $4M$ **($<0.5\%$ overhead)**
- Memory: 2.8M fp32 params $\approx$ **11 MB** vs. $\sim$2 GB base model
- LoRA correction runs on *digital* hardware — does not increase analog area

Bottom line

0.28% extra params, $<0.5\%$ extra FLOPs, correction entirely in digital post-processing. The trade-off is **strongly favourable**.

Thank you.

Questions?

Alina Burykina — Constructor University — April 2026